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Ethical Hacking

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Monday.

Co1 Assessment Tool-3

Independent Discovery of a Zero-Day Vulnerability

Introduction:

- A zero-day vulnerability is a previously unknown security weakness in software, hardware, or a computer system. It is called "zero-day" because the vendor has zero days to fix the vulnerability before it may be exploited

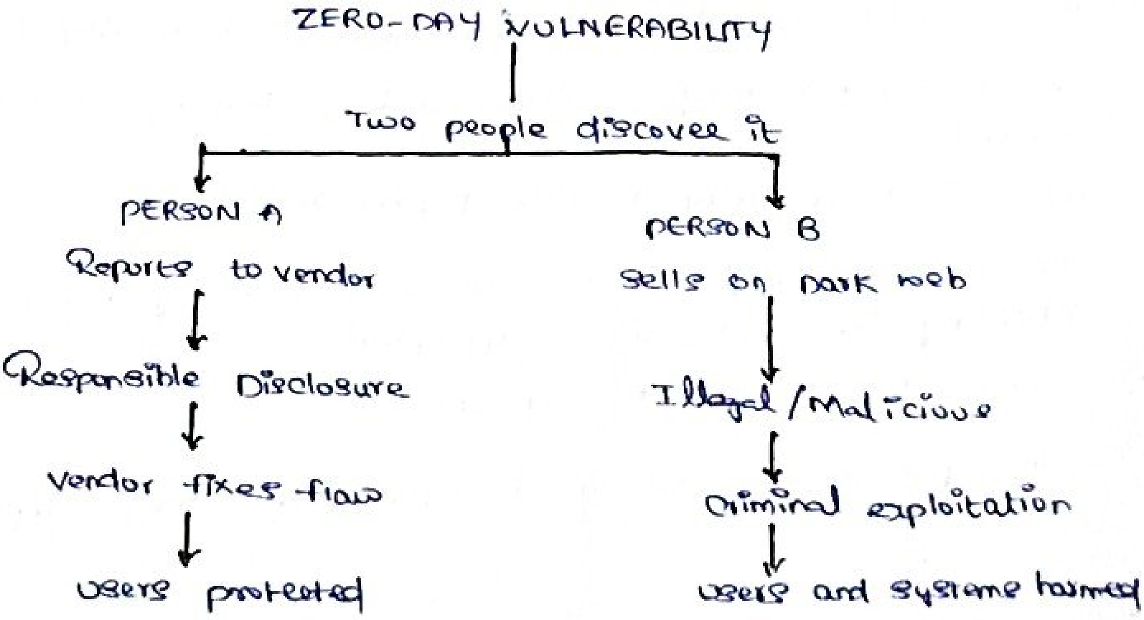
In this scenario, two people independently discover the same zero-day vulnerability

→ person A reports the vulnerability responsibly to the software vendor

→ person B sells the vulnerability through a dark web marketplace.

- Although both discovered the same technical weakness, their classification, motivation, ethical position, and consequences are very different.

1) Basic Concept



2) Person-A - Reports the vulnerability to the vendor.

Classification:

Person-A can be classified as an ethical hacker, white-hat hacker or security researcher, depending on the circumstances of the discovery

If the person reports the vulnerability through a proper vulnerability disclosure or bug bounty program, their actions are considered responsible disclosure.

Motivation

The main motivations may include:

→ protecting users and organizations

→ Helping the vendor fix the security weakness

→ Improving software security.

→ preventing criminals from exploiting the vulnerability.

→ Receiving recognition or a legitimate bug bounty reward.

→ Contributing to the cybersecurity community.

Consequences

The possible consequences for person A are generally positive.

→ The vendor can develop and release a security patch

→ The researcher may receive a bug bounty or financial reward

→ The researcher may receive public recognition

→ The vulnerability can be documented using a security advisory or CVE, where appropriate

→ users become better protected

→ The person's professional reputation may improve.

However, if the researcher accessed systems without authorization or violated laws while discovering the vulnerability they could still face legal issue. Responsible disclosure does not automatically make unauthorized access legal.

Concept Diagram

PERSON A
↓
Discovers zero-Day
↓
Reports to Software Vendor
↓
Responsible Disclosure
↓
Vendor Investigates
↓
Security patch Developed
↓
Users Receive protection
↓
positive security Impact

3) Person B - Sells the vulnerability on a Dark Web Marketplace

Classification:

person B may be classified as a black-hat actor or as someone participating in the criminal vulnerability market depending on the circumstances and intended use.

Selling a vulnerability is not automatically illegal in every jurisdiction, but selling it through a criminal marketplace especially for malicious exploitation can involve serious criminal activity

Motivation:

Possible motivations include:

- → Making money
- → Selling the vulnerability to attrackers
- → Supporting cybercrime
- → Seeking personal financial gain
- → Enabling espionage or targeted attacks.

Consequences

Person B may face serious consequences.

- → Criminal investigation and prosecution.
- → Financial penalties or imprisonment depending on applicable laws
- → Damage to personal and professional reputation.
- → The vulnerability may be exploited against innocent users.
- → Organizations may suffer data breaches and financial losses

Concept Diagram

PERSON B

↓

Discovers Zero-Day

↓

Sells Through Dark web

↓

Criminal Buyer

↓

Possible Cyber-Attack

Data Theft | System Damage

↓

Victim Harm

↓

Legal Investigation

4) Comparison Between the Two people

Factor	person A	person B.
Classification	Ethical (white-hat hacker or security researcher	Black-hat actor / criminal-market participant
Action	Reports vulnerability to vendor	sells vulnerability on dark web
Impact on users	Helps protect users	May expose users to attacks
Ethical position	Generally ethical	Generally unethical
Possible Reward	Bug bounty, recognition Career benefits	Illegal financial profit
Social Impact	Improves cybersecurity	Can increase cybercrime and security risks

5) Motivation Comparison

Same zero-Day

person A
↓
protect users
↓
Help Vendor
↓
Fix vulnerability
↓
Positive Impact

person B
↓
Earn money
↓
sell secret
↓
Enable exploitation
↓
Negative Impact

6) Consequences Comparsion

person A

Discovery

↓

Responsible Reporting

↓

vendor Notification

↓

security patch

↓

users protected

↓

possible Reward & Recognition

person B

Discovery

↓

Dark web sale

↓

potential criminal use

↓

cyber-Attacks

↓

Data / Financial Damage

↓

possible Law Enforcement Action

7) Ethical Analysis

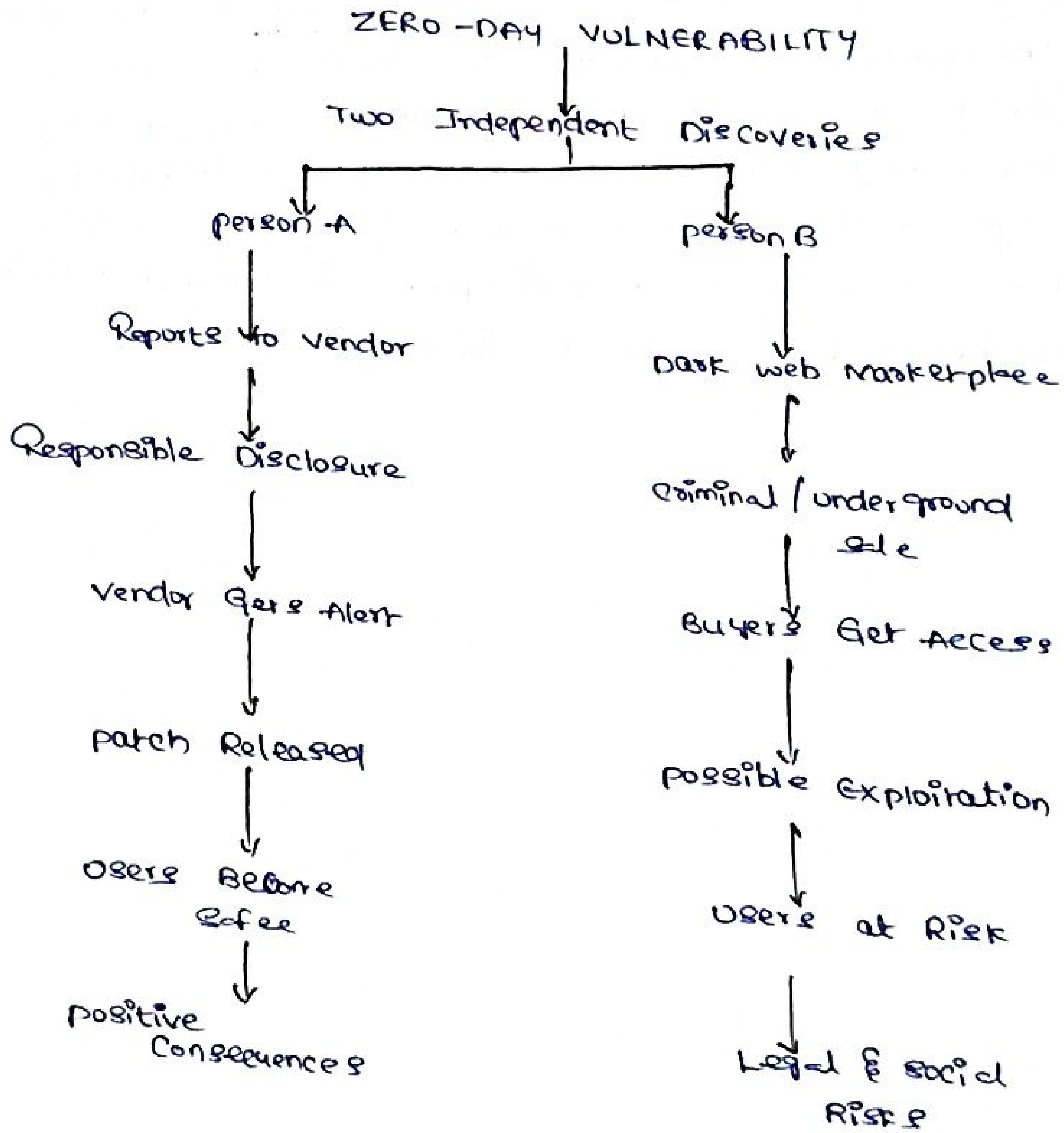
The important difference is not simply, who discovered the volunerability first. The major difference is what each person does after discovering it.

person A follows the principle of responsible disclosure. The person gives the vendor an oppootinity to understand and fix the vulnerability before it becomes widely exploited.

person B chooses to sell the vulnerability through an underground markerplace. This creates a risk that the vulnerability will be used to attack systems steal information or cause damage.

Therefore the ethical classification depends heavily on the intention method of disclosure authorization and consequences of the person's actions.

8) **Overall Concept Diagram**



Conclusion:

Both individuals discovered the same zero-day vulnerability but their actions lead to completely different outcomes person A who responsibly reports the Vulnerability to the vendor's is generally considered an ethical security researcher and contributes to improving cybersecurity. person B who sells the vulnerability through a dark web marketplace may be considered a black-hat actor or participant in criminal activity particularly if the sale facilities attacks.